

What does a released average-subject brain encoder actually predict? A noise ceiling, a parcel-level replication, and three silent failures

Brian Mwai^a

^a*Department of Natural Sciences, Faculty of Science, The Catholic University of Eastern Africa, P.O. Box 62157-00200, Nairobi, Kenya*

Abstract

Released encoding models are increasingly used as instruments: a checkpoint is downloaded, run on new stimuli, and its predictions treated as a proxy for cortical response. That use depends on a property the release does not certify, namely that the specific checkpoint, in the specific configuration its loader selects, predicts held-out cortex at all. A self-published audit reports that a widely used trimodal encoder, in the average-subject configuration its loader forces, is *anti-correlated* with cortex at vertex level, $r = -0.0145$ against an inter-subject ceiling of $+0.0508$. The claim is about sign rather than magnitude, comes from a company selling the per-subject alternative, and has not been independently replicated. We replicate it in parcel space on public data. From the Algonauts 2025 responses the leave-one-subject-out noise ceiling for the target episode is $r_{\text{ceiling}} = 0.1517$, 95% CI $[0.1484, 0.1551]$, over four subjects, all 1000 Schaefer parcels and 592 timepoints. Running the checkpoint over the first 120 s of that episode, with prediction and recording placed on one clock, gives a group mean $r = +0.0283$ against a ceiling of 0.0959 on the same 80 scans, $p = 0.048$ against a circular-shift null, with three of four subjects positive. The checkpoint is weakly positive and short of decisive, and it does not show the negative sign the audit reports. We also report three withdrawn values and the defects behind them, selecting a record by position, inferring orientation from axis length, and correlating a 2 Hz prediction index by index against a 1.49 s recording, because each is silent and each is reproducible by anyone attempting this measurement.

Email address: mwaibryn@gmail.com (Brian Mwai)

Keywords: encoding models, fMRI, noise ceiling, replication, model auditing

1. Introduction

A released encoding model invites a particular kind of use: treat the checkpoint as an instrument, run it on stimuli its authors never considered, and read its predictions as a stand-in for measured cortical response. The appeal is obvious, since the alternative is scanning people. The risk is equally obvious once stated, which is that a technical report describing an architecture is not a warranty about the weights released under its name, and that a checkpoint may be released in a configuration whose predictive validity nobody has established.

For one widely used trimodal encoder [1] that risk is not hypothetical. A commercial laboratory has published an audit of the released checkpoint in its average-subject configuration [2], reporting vertex-level $r = -0.0145$ against a measured inter-subject ceiling of $+0.0508$: anti-correlated with real cortex, across four subjects and all seven Yeo networks [5]. Taken at face value, that would mean every downstream study using this checkpoint as an instrument is reading a signal with no established relationship to cortex.

The audit should not be accepted uncritically. It is self-published by a company selling the per-subject alternative, it uses four subjects and one stimulus, and it reports magnitudes near 0.015 in either direction, so it is a claim about sign rather than about effect size. Neither should it be dismissed. It is specific, it is falsifiable, and it concerns a configuration that a released loader selects by default.

This paper replicates the audit’s question on public data, at parcel level, and reports the three silent defects this work met along the way.

2. The measurement that stands on its own

An encoder’s predictions are meaningless without knowing how much shared signal exists to predict. The leave-one-subject-out noise ceiling supplies that: each subject’s response is correlated against the mean of the others, giving a lower bound on what any encoder could achieve, computed from the recordings alone with no model involved.

2.1. Data and procedure

We use the publicly released Algonauts 2025 responses [3], four subjects viewing the same naturalistic episode, parcellated into 1000 Schaefer parcels [4] in MNI152NLin2009cAsym space. For each subject we correlate the parcel time series against the mean of the remaining three, per parcel, over time. Undefined correlations stay undefined: a parcel with no variance over time has no Pearson correlation, and substituting zero would pull the mean toward zero with invented values. Confidence intervals are percentile bootstrap over subjects with a fixed seed, so a reported interval reproduces exactly.

2.2. Result

For episode **s01e01a**, the episode a prediction pass would target,

$$r_{\text{ceiling}} = 0.1517, \quad 95\% \text{ CI } [0.1484, 0.1551], \quad (1)$$

over four subjects, 1000 of 1000 parcels defined, and 592 timepoints. Per-subject values are 0.1553, 0.1548, 0.1497 and 0.1470.

This is larger than the audit’s +0.0508 by construction rather than by disagreement. Averaging vertices within a parcel cancels independent noise and raises correlations, so a parcel-level ceiling and a vertex-level ceiling are not comparable numbers, and conflating them would manufacture an agreement or a disagreement that is really a change of spatial unit.

3. Three silent failures, and why they are reported

Three defects in this work changed a reported number without raising an error. Each is reported because any group repeating the measurement can reproduce it, and each was identified the same way: replaying the original code path on the same inputs reproduces the original number exactly, which establishes the cause rather than merely fitting it.

3.1. The ceiling: the wrong episode, transposed

An earlier version of the ceiling gave 0.2247, CI [0.2147, 0.2363], described as covering “482 parcels”. That value is withdrawn.

The episode was selected by position. The analysis code carried a placeholder for the stimulus identifier and fell back to the first key in the response file. That key is a different episode from the one under prediction, and subjects do not agree on which session an episode is stored under, so positional selection is inconsistent across files as well.

The array was transposed. The correlation routine expects (units, timepoints). The recordings are stored (timepoints, parcels), and the orientation guard tested whether the first axis was longer than the second. For a (482, 1000) array that test cannot fire, so every correlation ran across the parcel axis. The “482 parcels” in the withdrawn figure was the timepoint count. Replaying the original path reproduces 0.2247, its interval and the count of 482 exactly.

3.2. *The prediction: two clocks compared index by index*

The first completed prediction pass reported an encoder correlation of +0.0344, CI [+0.0198, +0.0433], against a ceiling of 0.1356 over 241 timepoints. That value is also withdrawn. The checkpoint emits one prediction per 0.5 s of stimulus: its loader cuts each segment at the model’s 2 Hz feature rate. The recordings are sampled once per 1.49 s [3]. The analysis truncated both arrays to the shorter length and correlated them sample by sample, so prediction k , at stimulus time $0.5k$ s, was scored against scan k , recorded at $1.49k$ s. By the last sample the two were four minutes apart. Replaying that path on the saved prediction reproduces +0.0344 and all four per-subject values exactly.

A synthetic check shows the size of the error. Two series drawn from one signal, one sampled as the prediction is and one as the recording is, correlate at $r > 0.999$ once aligned and at $r = -0.03$ when truncated and compared index by index.

Three practices follow, and we state them as recommendations because all three defects are generic. Select records by name, never by position. Decide array orientation against a known constant such as the atlas size, never by comparing axis lengths. And place prediction and recording on one clock, in seconds, before correlating anything: equal array lengths say nothing about equal sampling.

4. The prediction pass

4.1. *Procedure*

1. Load the released checkpoint [1] in the average-subject configuration its loader forces, which is the configuration the audit concerns.
2. Predict the response to the opening of episode `s01e01a`, the episode whose ceiling is reported above. The encoder is dataloader-bound, at

about 17 minutes of GPU time per 60 s of video on the hardware available, so a window of the episode is predicted rather than all of it.

3. Project predictions from the encoder’s fsaverage5 vertices into the 1000 Schaefer parcels [4], rather than upsampling the recordings, which would invent within-parcel structure the data does not have.
4. Map each recorded scan to the stimulus time it responds to. The checkpoint was trained against BOLD shifted back by 5 s, so a prediction for stimulus time t estimates the scan recorded at $t + 5$ s. The prediction is interpolated at $1.49j - 5$ s for each scan j whose stimulus time lies inside the predicted window; scans outside it are dropped, not extrapolated.
5. Measure the noise ceiling on exactly the scans used for the encoder, so the two numbers share one denominator.
6. Correlate per parcel over time and report the sign as measured. Nothing is flipped or absolute-valued.

The 5 s delay is taken from the checkpoint’s training configuration and fixed before any correlation was computed. Other delays are reported as sensitivity and are not selected from.

4.2. Inference

A bootstrap over four subjects has too few distinct resamples to carry an interval, so it is reported but not relied on. The test is a circular-shift null: the aligned prediction is rotated against the recordings by every shift of at least ten scans, which keeps each series’ autocorrelation intact and breaks only their alignment, and the group mean correlation is recomputed at each shift. The one-sided p is the share of shifts reaching the observed value, counting the observed alignment itself.

4.3. Result on the first 120 seconds

The first 120 s of the episode give 80 aligned scans, recording indices 4 to 83.

Three of four subjects are positive and one is indistinguishable from zero. Against the circular-shift null the group mean sits at the margin of chance, $p = 0.048$. Across response delays of 3 to 7 s the group mean stays positive, from $+0.0239$ to $+0.0377$, so the sign does not depend on the delay chosen. The ceiling on these 80 scans is lower than the full-episode 0.1517 because

Encoder, group mean r	+0.0283
Per subject (sub-01, 02, 03, 05)	−0.0048, +0.0316, +0.0541, +0.0322
Noise ceiling on the same scans	0.0959, CI [0.0885, 0.1025]
Circular-shift null, 61 shifts	mean −0.0048, sd 0.0191, max +0.0351
One-sided p	0.048

Table 1: The aligned prediction pass over the first 120 s of **s01e01a**, parcel level.

it is measured on fewer and earlier scans, which is why the two are never mixed.

Two minutes of film cannot settle the question. What they do establish is that on this window the released checkpoint is not anti-correlated with the recordings in parcel space: the one sign the audit predicts is the one the data here do not show.

5. Conclusion

The noise ceiling for the target episode is 0.1517, CI [0.1484, 0.1551], measured from public data and independent of any model. The aligned prediction pass over the first 120 s gives a group mean of +0.0283 against a ceiling of 0.0959 on the same scans, at $p = 0.048$ against a circular-shift null: positive, weak, and short of decisive. It does not reproduce the negative sign the audit reports, and at parcel level it is not comparable with the audit’s vertex-level −0.0145. Three withdrawn values and their causes are reported for the same reason a null is: each was silent, each is reproducible by anyone attempting this measurement, and none would have announced itself.

Code and data availability

The measurement script, the withdrawn code path, and the artifact recording every value above are public at <https://github.com/brn-mwai/monarch>. The ceiling is produced by `scripts/paper3_noise_ceiling.py`, which takes the episode as an argument, matches the record by name and asserts orientation against the atlas size. The prediction pass is `scripts/paper3_prediction.py`, which accepts a saved prediction so the analysis can be rerun without a GPU; the alignment is `app/services/temporal_alignment.py`; the null and the delay sensitivity are `scripts/paper3_alignment_checks.py`.

CRedit authorship contribution statement

Brian Mwai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review and editing, Visualization. The author is the sole contributor to this work.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. Computation was carried out on publicly available free-tier resources.

Acknowledgements

The author thanks Dr. Songa Mutambi of the Department of Natural Sciences, The Catholic University of Eastern Africa, for supervising the project of which this work forms part. The supervision was academic guidance and approval; the author is solely responsible for the content.

References

- [1] S. d’Ascoli, J. Rapin, Y. Benchetrit, H. Banville, J.-R. King, TRIBE: TRImodal Brain Encoder for whole-brain fMRI response prediction, arXiv:2507.22229 (2025).
- [2] The Sapien Company, The Faceless Brain, self-published technical report, 2026, <https://thesapiencompany.com/research>.
- [3] A. T. Gifford, D. Bersch, M. St-Laurent, et al., The Algonauts Project 2025 Challenge: How the Human Brain Makes Sense of Multimodal Movies, arXiv:2501.00504 (2025).

- [4] A. Schaefer, R. Kong, E. M. Gordon, T. O. Laumann, X.-N. Zuo, A. J. Holmes, S. B. Eickhoff, B. T. T. Yeo, Local-global parcellation of the human cerebral cortex from intrinsic functional connectivity MRI, *Cereb. Cortex* 28 (2018) 3095–3114.
- [5] B. T. T. Yeo et al., The organization of the human cerebral cortex estimated by intrinsic functional connectivity, *J. Neurophysiol.* 106 (2011) 1125–1165.